



Analytic Continuation & SmoQyDEAC.jl

UC Davis

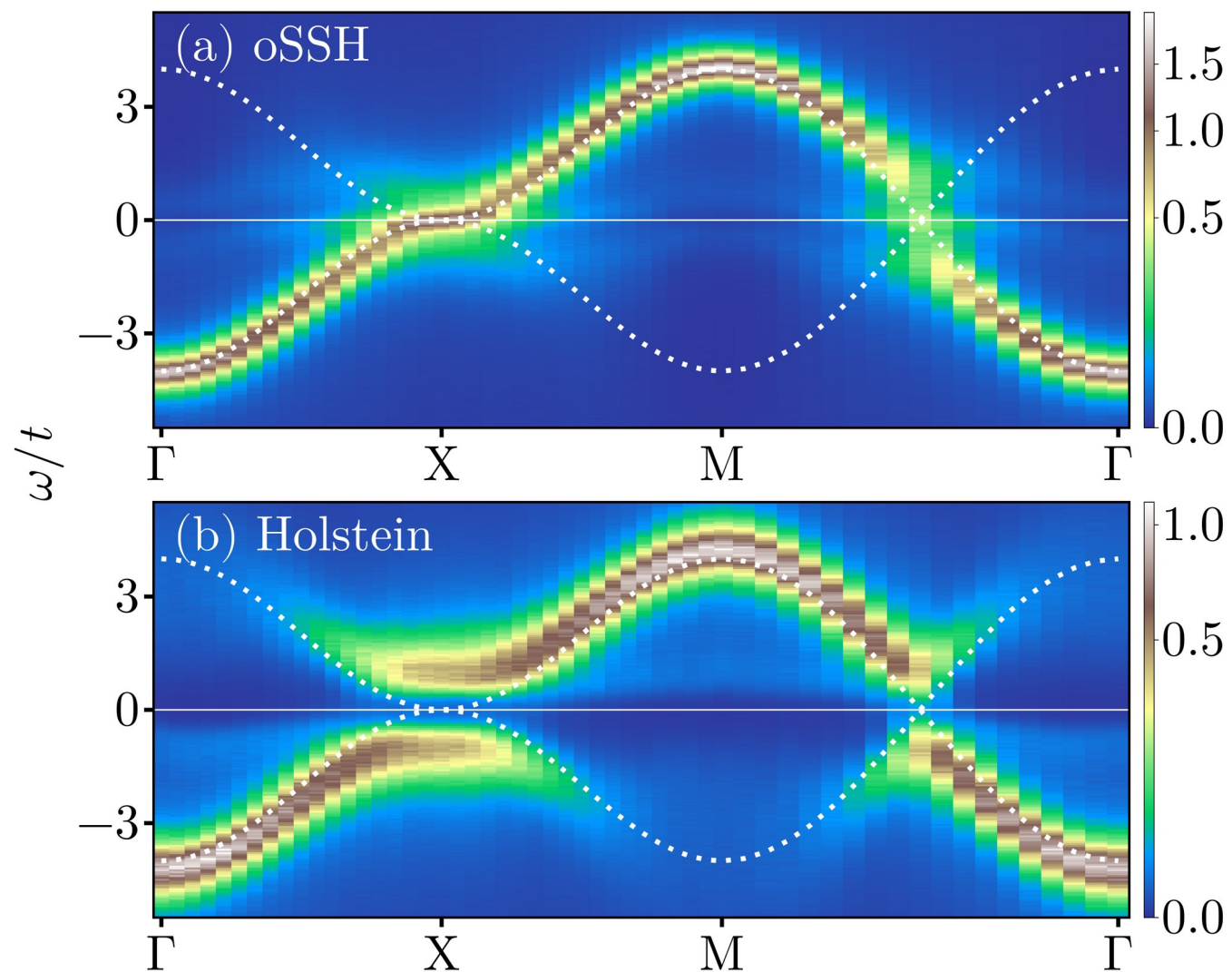


THE UNIVERSITY OF
TENNESSEE
SYSTEM

About me:



Amuse Bouche



Agenda

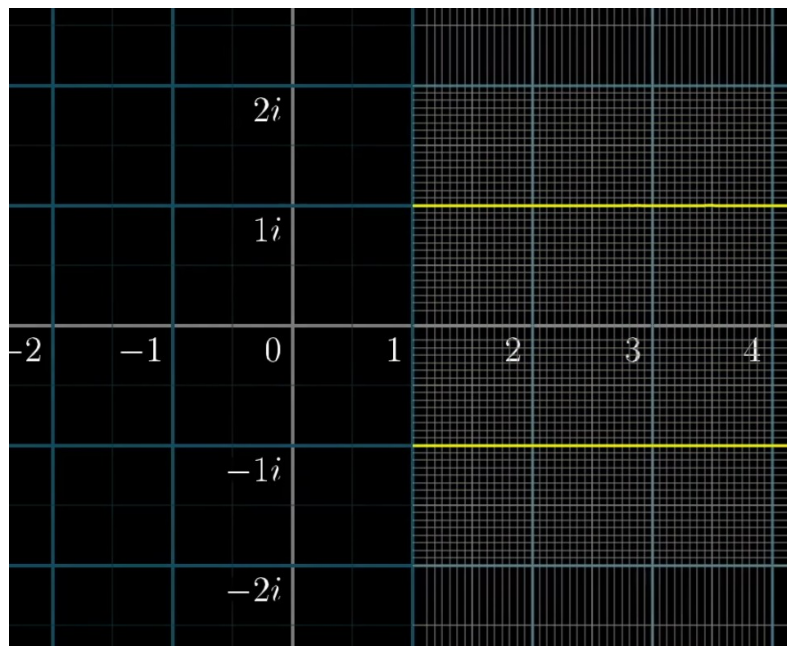
- What is Analytic Continuation
- MaxEnt Primer
- Other Methods
- DEAC algorithm
- Using SmoQyDEAC.jl

1. What is Analytic Continuation?

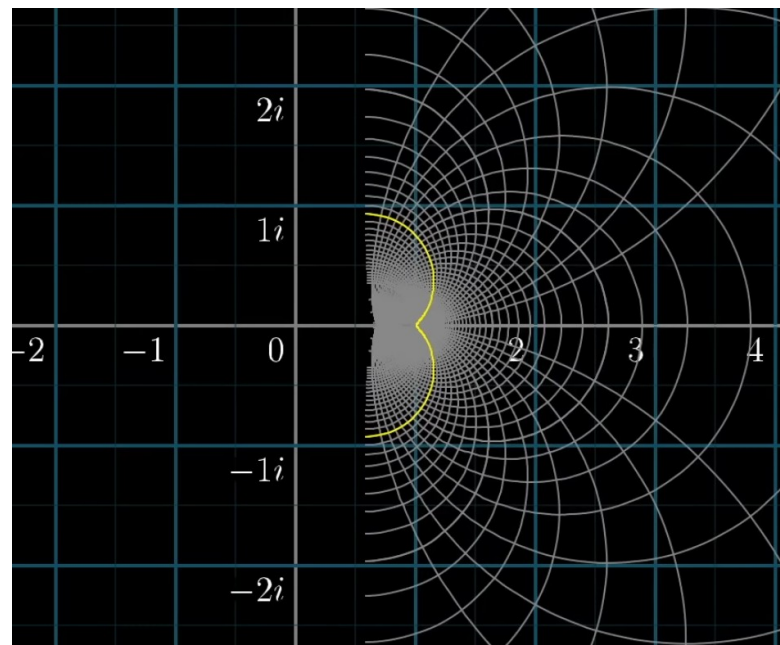
- Extends complex analytic function from one domain to larger domain
- An analytic function can be written as $f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$ for some neighborhood $r > 0, |z - z_0| < r$
- For complex functions this means they are differentiable everywhere they are defined
- New mapped function is meromorphic and **UNIQUELY DEFINED***
 - Meromorphic: analytic everywhere except may contain poles

Euler Zeta Function

$$\Re(z) > 1$$



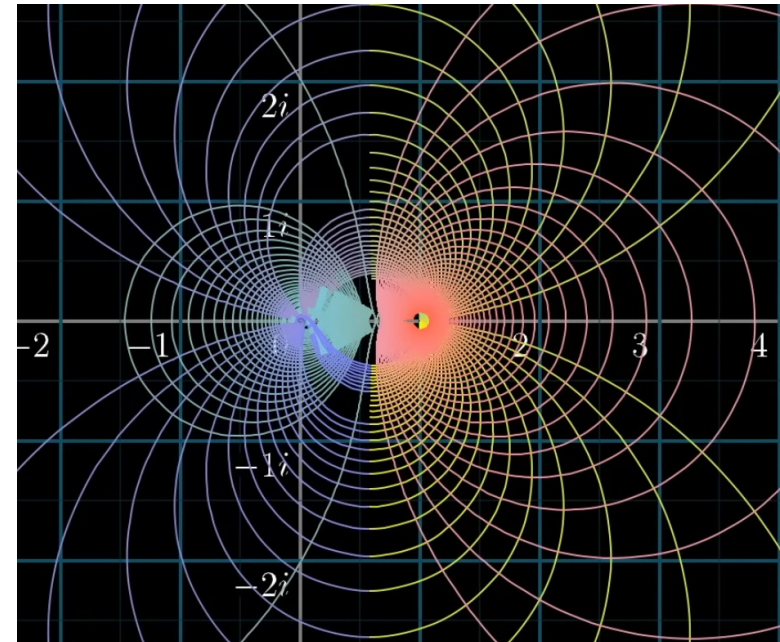
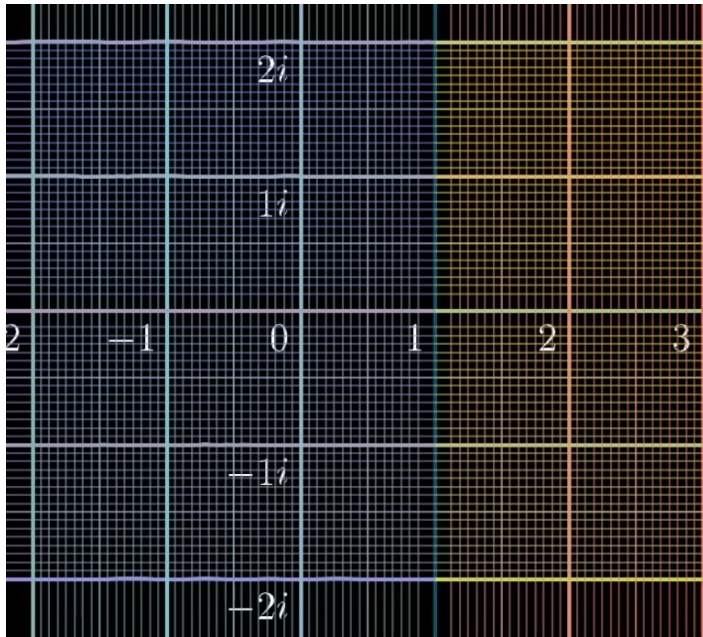
$$\zeta(z) = \sum_{n=1}^{\infty} n^{-z}$$



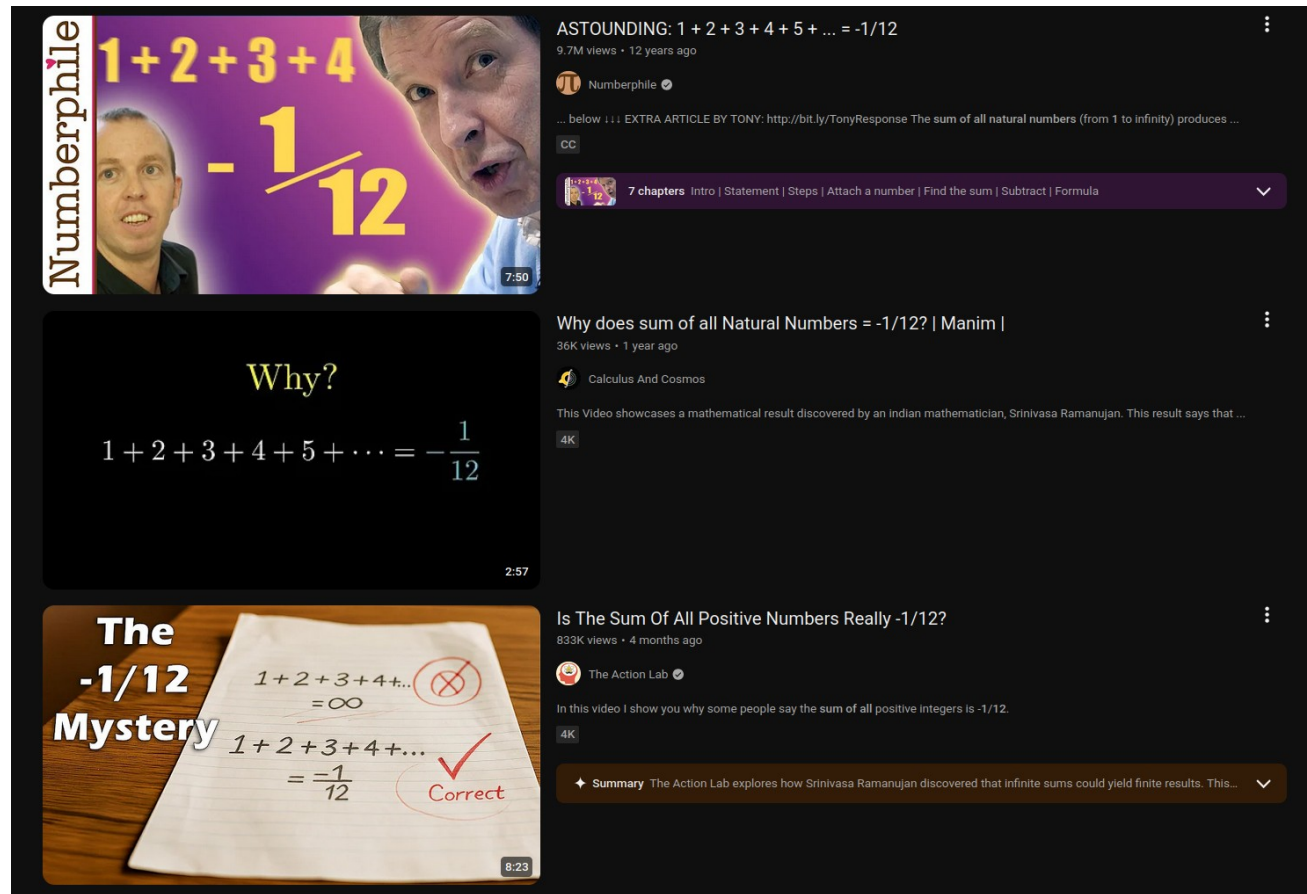
Riemann Zeta Function

z

$$\zeta(z) = \sum_{n=1}^{\infty} n^{-z}$$



YouTube “Math”



The screenshot shows a YouTube search results page for the Ramanujan summation. The top result is from the channel 'Numberphile', titled 'ASTOUNDING: $1 + 2 + 3 + 4 + 5 + \dots = -1/12$ ', with 9.7M views and posted 12 years ago. The video thumbnail features the Numberphile logo, the equation $1+2+3+4 - 1/12$, and two men. Below the video player, there is a section titled 'Why?' showing the equation $1 + 2 + 3 + 4 + 5 + \dots = -\frac{1}{12}$. The second result is from 'Calculus And Cosmos', titled 'Why does sum of all Natural Numbers = -1/12? | Manim |', with 36K views and posted 1 year ago. The third result is from 'The Action Lab', titled 'Is The Sum Of All Positive Numbers Really -1/12?', with 833K views and posted 4 months ago. The video thumbnail for this result shows a piece of paper with the equation $1 + 2 + 3 + 4 + \dots = \infty$ crossed out and $1 + 2 + 3 + 4 + \dots = -\frac{1}{12}$ marked as 'Correct' with a red checkmark.

Numberphile

ASTOUNDING: $1 + 2 + 3 + 4 + 5 + \dots = -1/12$

9.7M views • 12 years ago

Numberphile

... below : EXTRA ARTICLE BY TONY: <http://bit.ly/TonyResponse> The sum of all natural numbers (from 1 to infinity) produces ...

CC

7 chapters Intro | Statement | Steps | Attach a number | Find the sum | Subtract | Formula

Why?

$1 + 2 + 3 + 4 + 5 + \dots = -\frac{1}{12}$

2:57

The

-1/12

Mystery

$1 + 2 + 3 + 4 + \dots = \infty$

$1 + 2 + 3 + 4 + \dots = -\frac{1}{12}$

Correct

8:23

Why does sum of all Natural Numbers = -1/12? | Manim |

36K views • 1 year ago

Calculus And Cosmos

This Video showcases a mathematical result discovered by an Indian mathematician, Srinivasa Ramanujan. This result says that ...

4K

Is The Sum Of All Positive Numbers Really -1/12?

833K views • 4 months ago

The Action Lab

In this video I show you why some people say the sum of all positive integers is -1/12.

4K

Summary The Action Lab explores how Srinivasa Ramanujan discovered that infinite sums could yield finite results. This...

We aren't so lucky

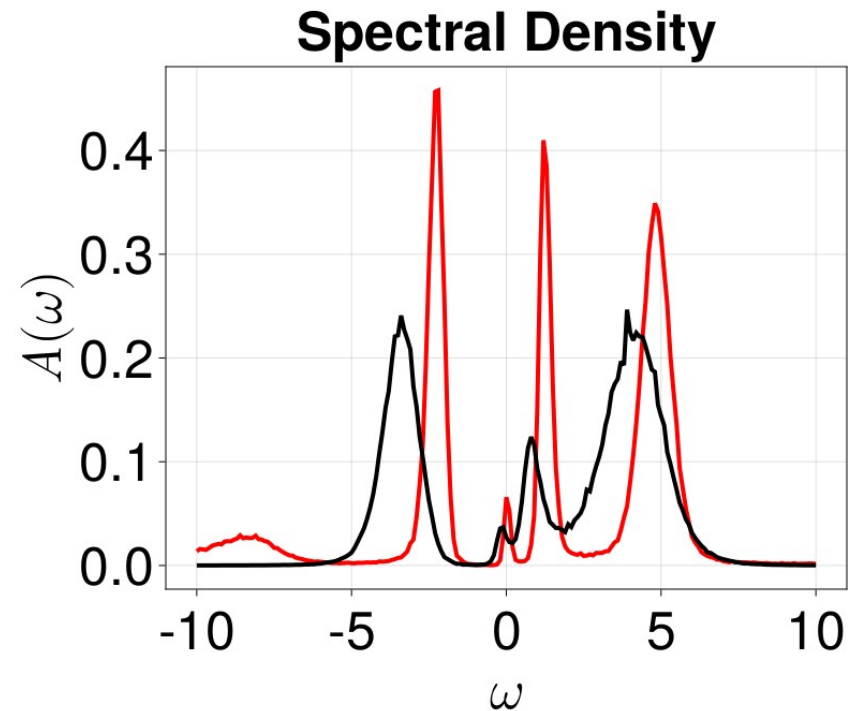
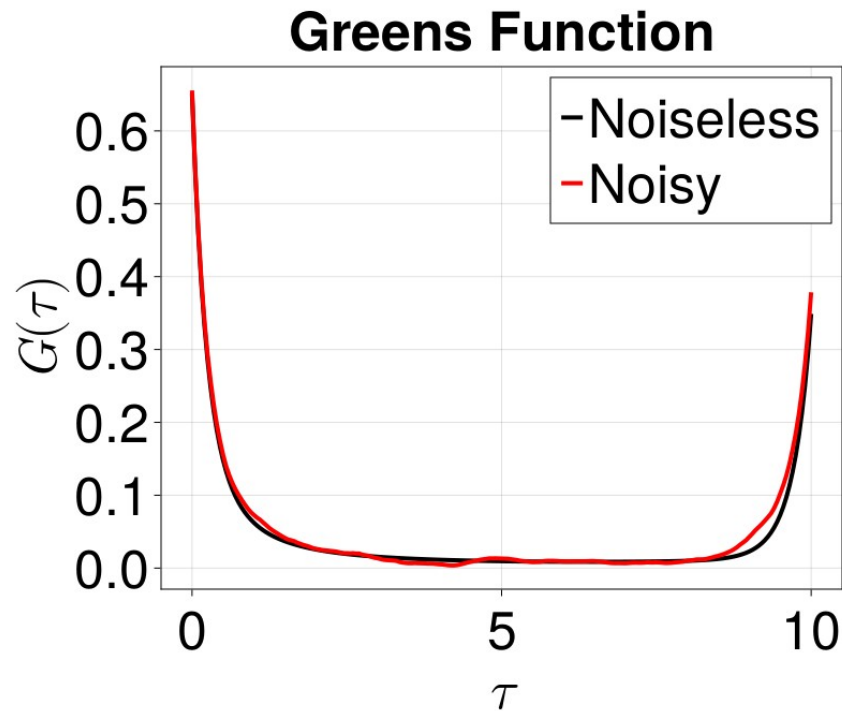
- Retardation means Green's functions are analytic on the upper half of the complex plane, but...
- QMC only takes measurements at discrete points on imaginary axis!
- QMC is also noisy
- Using just MC output we can no longer ensure uniqueness
- Naive attempts at AC will give garbage results

AC on QMC

$$G(i\omega_n) = \int_{-\infty}^{\infty} d\omega \frac{1}{i\omega_n - \omega} A(\omega)$$

$$G(\tau) = \int_{-\infty}^{\infty} d\omega \frac{e^{-\tau\omega}}{1 \pm e^{-\beta\omega}} A(\omega)$$

$$\mathbf{G} = \mathbf{K}\mathbf{A}$$



2. MaxEnt

M. Jarrell and J. Gubernatis, Bayesian inference and the analytic continuation of imaginary-time quantum Monte Carlo data, *Physics Reports* **269**, 133 (1996).

- Utilizes “model” and entropy term to fight overfitting to noise
- Given α , maximize posterior likelihood Q

$$Q = \alpha S - \frac{\chi^2}{2}$$

Entropy

$$S = \int d\omega \left(A(\omega) - M(\omega) - A(\omega) \log \left[\frac{A(\omega)}{M(\omega)} \right] \right)$$

$$S = \sum_i \Delta\omega \left(A(\omega_i) - M(\omega_i) - A(\omega_i) \log \left[\frac{A(\omega_i)}{M(\omega_i)} \right] \right)$$

Fitness

$$\chi^2 = \sum_{i,j}^L (\bar{G}_i - K_{ik} A_k) [C^{-1}]_{ij} (\bar{G}_j - K_{jk} A_k)$$

$$\chi^2 \approx \sum_i^L \frac{(\bar{G}_i - K_{ik} A_k)^2}{\sigma_i^2}$$

2. MaxEnt (cont)

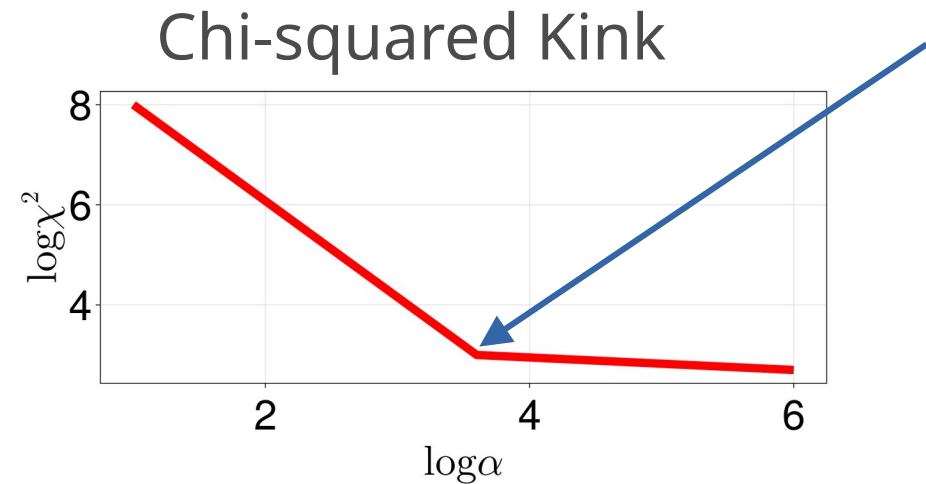
- User inputs array of alphas and default model M
- For each alpha MaxEnt uses solver to find A which maximizes Q
- Based on results for each alpha value, A is determined (next slide)

MaxEnt (cont)

- 2 Free parameters, $M(\omega)$ and α
- Result is typically not terribly reliant on choice of $M(\omega)$ (YMMV)
 - I usually just use a wide Gaussian centered on 0
- Alpha needs to be found

Bryan Algorithm

$$\bar{A}(\omega) = \int d\alpha A_{\alpha}(\omega) P[\alpha | \bar{G}]$$



ana_cont

- My go-to MaxEnt code
- Python based
 - I hate Python, so I run it in Julia using PyCall without issue
- Can utilize full covariance matrix!
- Example Python and Julia scripts in GitHub repo

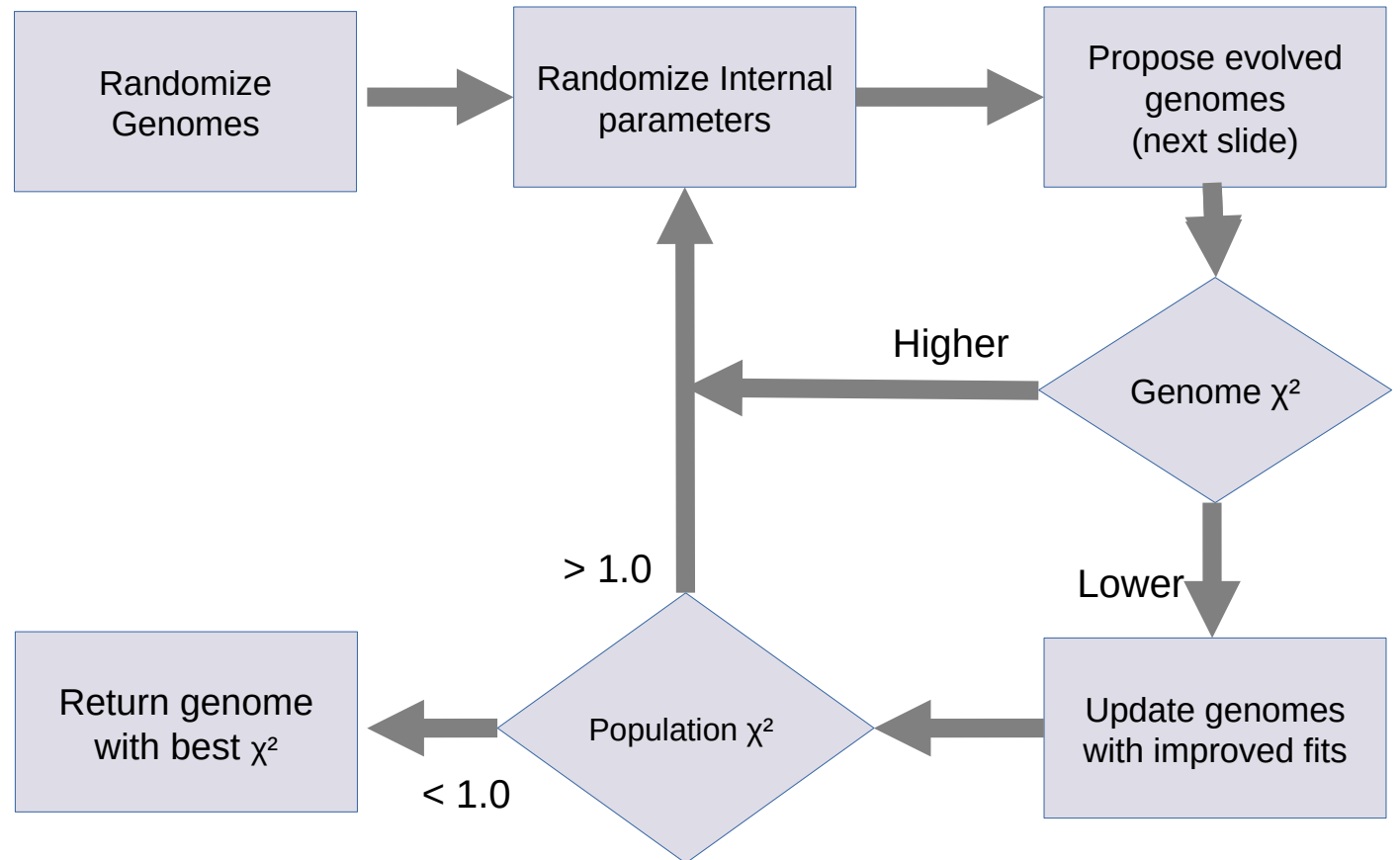
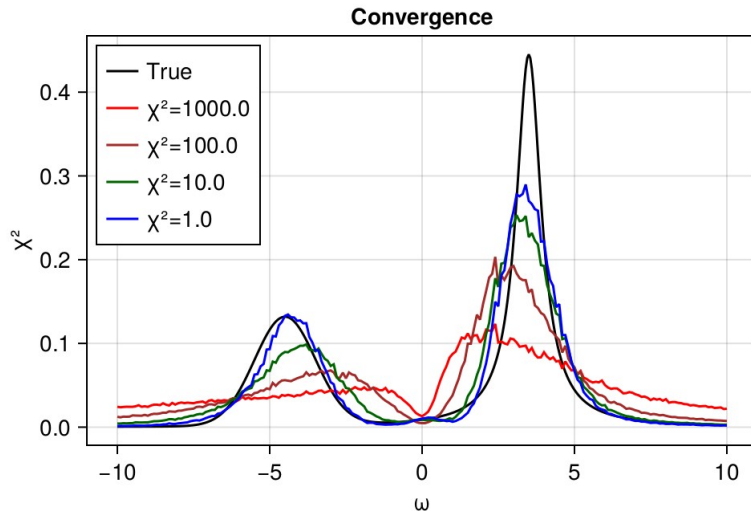
3. The AC Method Menagerie

- Nevanlinna – Not Noise resistant
- StochasticAC – Sandvik group: publishes papers on method but not code
- StochasticOM – I haven't found a good, usable code
- PronyAC – Emanuel Gull's group, not noise resistant
- Padé Approximation – Not Noise resistant
- Machine Learning methods – **Nobody** I trust to solve the problem is even working on this

4. Differential Evolution for AC

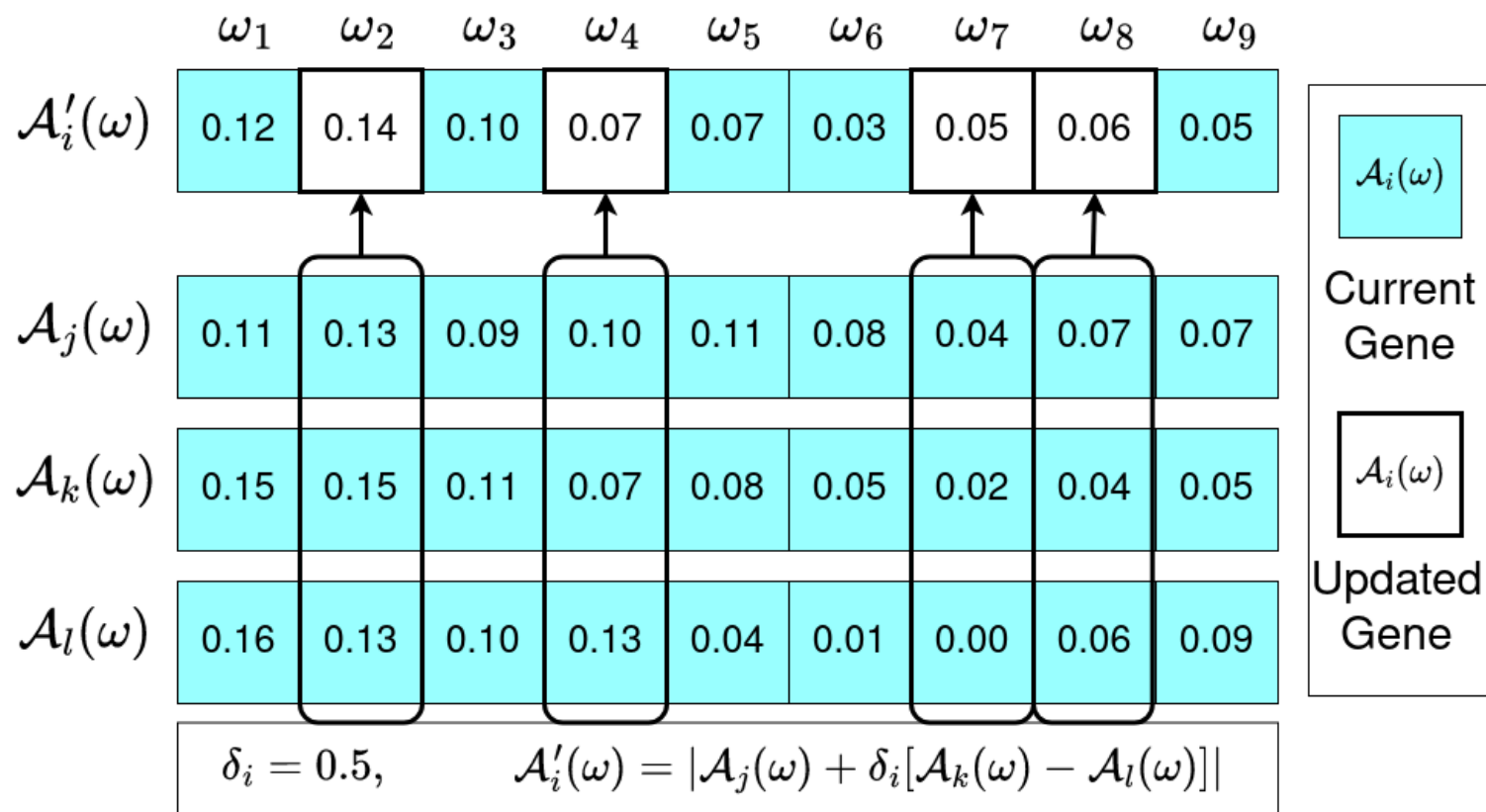
$$G_{\text{Calc}}(\tau_\alpha) = K(\tau_\alpha, \omega) A(\omega)$$

$$\chi^2 = \frac{1}{N} \sum_{\alpha=1}^{N=\beta/\Delta\tau} \frac{(G_{\text{Meas}}(\tau_\alpha) - G_{\text{Calc}}(\tau_\alpha))^2}{\sigma_i^2}$$



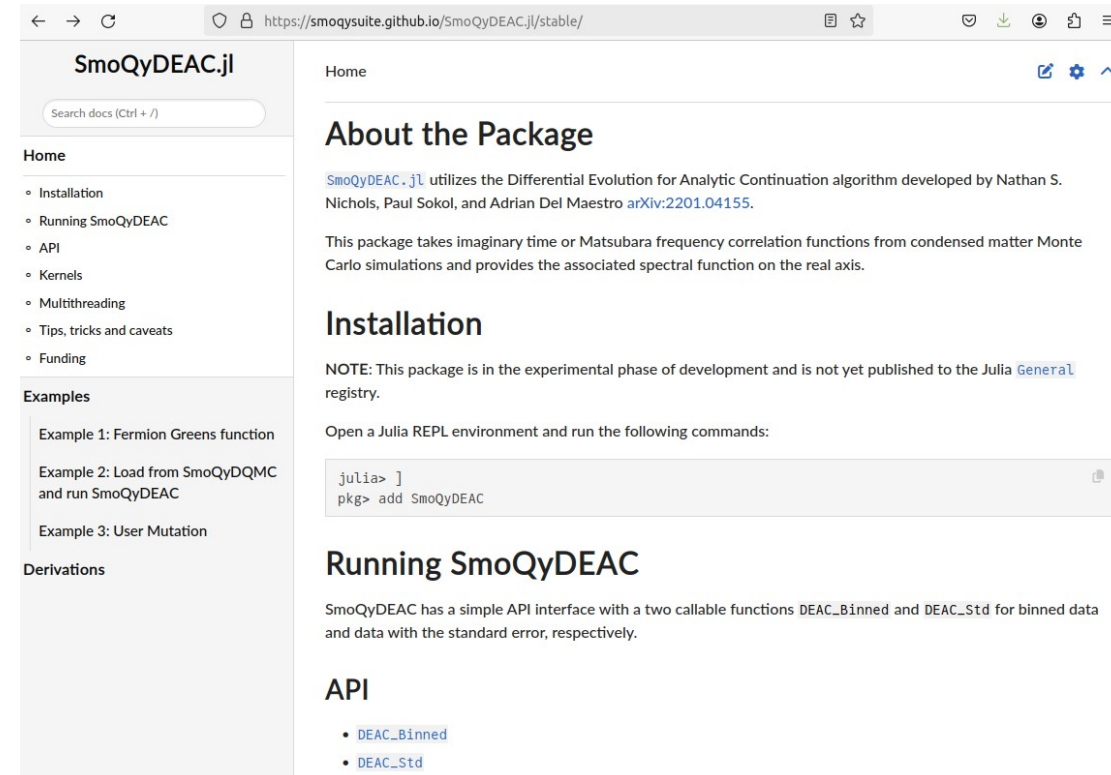
N Nichols et al. Phys. Rev. E 106, 025312 (2022)

DEAC updates



SmoQyDEAC.jl

- Allows Multiple χ^2 targets
- Fitness Floor Finder
- User defined mutations
- Multi-threaded, SIMD accelerated



The screenshot shows the documentation page for SmoQyDEAC.jl, hosted on GitHub. The page is titled "SmoQyDEAC.jl" and has a search bar at the top. The left sidebar contains a navigation menu with links to "Home", "Installation", "Running SmoQyDEAC", "API", "Kernels", "Multithreading", "Tips, tricks and caveats", "Funding", "Examples", and "Derivations". The main content area is titled "About the Package" and describes the package's purpose: "SmoQyDEAC.jl utilizes the Differential Evolution for Analytic Continuation algorithm developed by Nathan S. Nichols, Paul Sokol, and Adrian Del Maestro arXiv:2201.04155." It also mentions that the package takes imaginary time or Matsubara frequency correlation functions from condensed matter Monte Carlo simulations and provides the associated spectral function on the real axis. Below this, there is an "Installation" section with a note that the package is in the experimental phase of development and is not yet published to the Julia General registry. It provides instructions to open a Julia REPL environment and run the following commands:

```
julia> ]
pkg> add SmoQyDEAC
```

 The "Running SmoQyDEAC" section states that the package has a simple API interface with two callable functions: `DEAC_Binned` and `DEAC_Std` for binned data and data with the standard error, respectively. The "API" section lists these two functions.

Using ana_cont and SmoQyDEAC.jl

